Krish Maniyar

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Education

Vellore Institute of Technology (VIT) **2023 – 2027**
B.Tech, Computer Science and Engineering | ***CGPA: 9.27 / 10*** *Vellore, Tamil Nadu*

Experience

Marwadi Shares & Finance Ltd. (MSFL) **May 2025 – Jun 2025**
Flutter Application Development Intern *Rajkot, Gujarat*

- Developed the Flutter mobile frontend for **MChat**, a real-time chat application serving **150+ organizational users** (supporting **150+ concurrent connections**), integrating a .NET Web API and a **SignalR hub** for group/direct chats, read receipts, and presence tracking, reducing message latency by **40%** (verified by QA testing).
- Engineered **reliable SignalR connection management** with automatic reconnection, heartbeat/keep-alive pings, app pause/resume lifecycle hooks, and a **Long Polling fallback** to resolve WebSocket negotiation failures in the UAT environment.
- Built a secure **file management system** supporting multipart uploads up to **5 GB**, background download queues, and local gallery storage with platform permissions; debugged server-side SSL `CERTIFICATE_VERIFY_FAILED` errors caused by IP mismatches.

Centre for Social Entrepreneurship and Development (CSED), VIT **Apr 2024 – Feb 2025**
Core Committee Member *Vellore, Tamil Nadu*

- Organized a **national-level hackathon** with **200+ participants**, coordinating event logistics, sponsor engagement, and cross-functional teams, achieving a **60% increase** in event reach.
- Collaborated with student clubs, faculty, and external partners to plan and execute technical and entrepreneurship-focused events across the academic year.

Projects

Strategic News Analyzer | *FastAPI, PostgreSQL, pgvector, Redis, Celery, Next.js, Docker, RAG* GitHub

- Built a **24/7 news ingestion pipeline** (FastAPI + Celery Beat) pulling from **5+ sources**, deduplicating articles via **SHA-256 hashing**, auto-translating non-English content, and running **parallel AI analysis** (sentiment, bias, strategic scoring) — processing ~**400 articles/hour**.
- Designed a **hybrid RAG system** using **pgvector** semantic search and **PostgreSQL full-text search**, streaming AI responses over **SSE**; extracted entity **knowledge graphs** and generated **intelligence forecasts** with confidence scoring.
- Containerized and deployed the full backend stack on **AWS EC2** via **Docker Compose**; served the Next.js frontend on **Vercel** and used **Supabase** as the managed database layer with pgvector, JWT auth, and Row-Level Security.

Query Weaver | *FastAPI, React, MySQL, ChromaDB, SQLAlchemy, Groq LLM, Sentence-Transformers, SSE* GitHub

- Built a **natural-language-to-SQL system** embedding user questions via `all-MiniLM-L6-v2` and performing **vector similarity search** in ChromaDB to retrieve the top-3 relevant tables; injected schema context, sample rows, and conversation history into **Groq (Llama 3.3-70B)** for SQL generation (end-to-end latency: ~**1.14s**).
- Implemented **SQL safety controls** detecting `INSERT`/`UPDATE`/`DELETE`/DDL and requiring explicit confirmation before execution; **auto-synchronizes schema changes** to ChromaDB to keep the vector store consistent with the live database.
- Streamed schema-change events to the React SPA via **Server-Sent Events**; instrumented per-stage timing — schema retrieval **44ms**, SQL execution **2–3ms**, LLM generation ~**1.02s** — via structured rotating logs.

Aneri Enterprise | *Next.js, TypeScript, Supabase (PostgreSQL), Nodemailer, Netlify* anerienterprise.shop

- Built and **deployed to production** a full-stack business website with a dynamic product catalog, secure driver download portal, and dual-channel enquiry system (Supabase DB + Gmail SMTP) serving real customers.
- Hardened the API with **IP-keyed rate limiting**, honeypot anti-spam, XSS/CRLF injection prevention, and an **SSRF-protected download proxy**; enforced **PostgreSQL RLS** restricting the anon key to SELECT on content tables and INSERT on analytics tables only.
- Built a **fire-and-forget analytics system** tracking site visits, product clicks, and driver downloads — all stored in Supabase with insert-only RLS; implemented **parallel data fetching**, session-based deduplication, and a 280ms debounced search to minimize redundant renders and DB writes.

Technical Skills

Languages: Python, SQL, C++, Java

Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, Statistical Modeling, Predictive Analytics, Regression (Linear/Logistic, Ridge/Lasso), Random Forests, Gradient Boosting, SVM, Clustering (K-Means)

Generative AI & Deep Learning: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), PyTorch, Hugging Face Transformers, LangChain, Sentence-Transformers, NLP

Visualization & Web Frameworks: Streamlit, Plotly, Dash, Matplotlib, Seaborn, FastAPI, Flask

Cloud, Data Warehousing & MLOps: Google Cloud Platform (GCP), AWS (EC2, S3), BigQuery, Docker, Apache Airflow, MLflow, Git/GitHub

Databases & Vector Search: PostgreSQL, pgvector, MySQL, ChromaDB, Redis, Supabase

Certifications & Achievements

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate **Jan 2026**
Oracle University